

llmXive Automated Review of Foundation Protocol: A Coordination Layer for Agentic Society

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is advisory, automated feedback for the authors: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer_claim_accuracy

Verdict: minor_revision

The manuscript makes several strong claims regarding the historical progression of industrial revolutions and the current state of agent protocol literature, supported by citations that require closer scrutiny for direct alignment.

In Section 1.1, the narrative of industrial revolutions as a linear progression of "intelligence density" is supported by Schwab (2016) and Hermann (2016). However, the specific claim that Industry 4.0 "fused networks, sensors, and cyber-physical feedback loops" is presented as a historical fact. While these sources define Industry 4.0, they may not explicitly frame this "fusion" as the definitive historical signature in the same way the paper does for previous revolutions. The citation supports the *existence* of Industry 4.0 concepts, but the specific historical narrative of "fusion" as a completed step is an interpretation by the authors. This is a minor overstatement of the direct evidence provided by the cited sources.

In Section 1.2, the paper relies on two very recent preprints (Tomašev et al., 2025; Catalini et al., 2026) to support the claim that "verification capacity, cryptographic provenance, and liability underwriting" are the specific scarce complements in the emerging economy of autonomous agents. While these papers likely discuss these themes, the claim is presented as a settled economic reality ("Recent economic analyses make this pressure clear"). Given the preprint status and the specific, strong nature of the claim, the authors should ensure the citations explicitly support this precise triad of scarce resources, or soften the language to indicate this is a hypothesis derived from emerging literature.

In Section 1.3, the claim that “Recent surveys of agent protocols point to related gaps around collaboration, scalability, security, privacy, and group-based interaction” cites a single 2025 survey (Yang et al.). It is crucial to verify that this specific survey explicitly enumerates *all* these gaps together. If the survey covers some but not all, or if the authors are synthesizing these from multiple sources, the single citation is insufficient to support the comprehensive list. The claim risks overstating the direct support of the single source.

Finally, in Section 3.2, the paper attributes the phrase “untrusted code execution with durable privileges” to Microsoft’s security research team. If this is a direct quote, the citation is appropriate. If it is a paraphrase or summary by the authors, the phrasing should be adjusted to avoid implying the source used these exact words, which could be a misrepresentation of the original text.

Overall, the claims are generally well-grounded in the cited literature, but in several instances, the strength of the claim slightly exceeds the direct, explicit support of the specific citations, particularly regarding the precise wording of historical narratives and the comprehensive nature of survey findings.

- [writing] Section 1.1 claims Industry 4.0 ‘fused networks...’ citing Schwab/Hermann. These define the concept but may not explicitly frame this ‘fusion’ as a completed historical signature. Clarify if this is an author interpretation or a direct finding.
- [writing] Section 1.2 cites 2025/2026 preprints to claim verification/provenance are the ‘primary scarce complements.’ Ensure these sources explicitly support this specific triad, or soften the claim to reflect emerging hypothesis.
- [writing] Section 1.3 cites Yang (2025) for a specific list of gaps (collaboration, scalability, etc.). Verify the survey explicitly enumerates all these gaps together; otherwise, the single citation overstates support for the comprehensive list.
- [writing] Section 3.2 attributes ‘untrusted code execution with durable privileges’ to Microsoft. If this is a paraphrase, soften phrasing to avoid misrepresenting the source’s exact terminology.

paper_reviewer_figure_critic

Verdict: accept

Figure 1

Figure 1 is a clear, well-structured infographic that effectively visualizes the evolution of the web from 1.0 to 4.0. The layout is uncluttered, the text is legible, and the content aligns perfectly with the caption’s description of raising capability and exposing coordination failures.

Figure 2

Figure 2 effectively visualizes the Foundation Protocol architecture with four distinct planes and a configuration/profile plane, matching the caption’s description. The diagram is clear, well-organized, and uses consistent visual cues to represent the relationships

between components.

paper_reviewer_jargon_police

Verdict: minor_revision

The manuscript relies heavily on specialized terminology from distributed systems, cryptography, and software architecture, often without providing plain-English definitions or context for a broader audience. While the paper aims to describe a coordination layer for an "agentic society," the density of jargon creates a barrier to entry for readers who are not experts in protocol design or computer science.

Specific instances of overuse include the repeated use of "first-class" (Sections 1.1, 3.1, 3.3) to describe protocol objects. This is a programming metaphor that should be replaced with "explicit," "native," or "integral" to improve clarity. Similarly, "progressive disclosure" (Section 1.4) is introduced as a design constraint without explaining the mechanism (exchanging minimal metadata first) to the general reader.

Technical terms like "backpressure" (Section 3.3), "ledger-agnostic" (Section 1.4), and "behavioral closure" (Section 1.4) are used as if they are common knowledge. "Backpressure" is a specific streaming concept that needs a brief explanation of how it prevents system overload. "Behavioral closure" is a formal logic term that obscures the simple idea of "completing the set of necessary actions."

Furthermore, the paper frequently uses "envelope" (Section 3.1) and "checkpoint" (Section 3.5) as metaphors for message wrappers and policy gates. While defined in tables, these terms appear in the main text without immediate clarification, forcing the reader to cross-reference tables to understand the prose. The phrase "flag-day migration" (Section 1.4) is also a niche industry term that should be simplified to "sudden, system-wide change."

To make the paper accessible to the intended "agentic society" audience—which includes policymakers, business leaders, and non-specialist researchers—the authors should replace these jargon-heavy phrases with descriptive, plain-language equivalents or provide immediate, concise definitions upon first use.

- [writing] The manuscript relies heavily on specialized terminology from distributed systems, cryptography, and software architecture, often without providing plain-English definitions or context for a broader audience. While the paper aims to describe a coordination layer for an "agentic society," the density of jargon creates a barrier to entry for readers who are not experts in protocol design or computer science. Specific instances of overuse include the repeated use of "first-class" (Sections 1.1, 3.1

paper_reviewer_logical_consistency

Verdict: minor_revision

The manuscript presents a coherent high-level architecture for the Foundation Protocol (FP), and the logical flow from the problem statement (fragmentation in agentic coordina-

tion) to the proposed solution (a graph-native, plane-based protocol) is generally sound. The distinction between the protocol core and the implementation profiles is logically consistent throughout Sections 1 and 2.

However, there are specific areas where the logical connection between premises and conclusions requires tightening:

1. **Syntactic Logic in Enumerations:** In Section 2.3 ("Interaction & Organization Plane"), the text lists the primitives provided by the plane: "Schemas define... Events and streams provide... Sessions and organizations capture... and Economic primitives standardize...". The inclusion of the conjunction "and" at the start of the final clause creates a sentence fragment that disrupts the logical parallelism of the list. While a minor writing issue, it obscures the logical grouping of the four distinct primitives the authors claim the plane provides.

2. **Mechanism for Asset Accountability:** In Section 2.1 ("Entity & Trust Plane"), the authors premise that the unified entity model allows organizations to "hold assets" and "act as counterparties." The conclusion drawn is that this enables "accountability" and "governance." However, the paper simultaneously asserts in Section 2.4 and the Appendix that FP is "ledger-agnostic" and does not mandate a payment rail. The logical gap lies in the mechanism: if the protocol does not manage the ledger, how does the "Entity" model enforce the "holding" of assets or prevent double-spending in a way that satisfies the "accountability" claim? The paper asserts that receipts and settlement references provide this, but it does not logically bridge the gap between a *reference* to an external payment and the *enforcement* of asset ownership within the protocol's graph. The argument assumes that "referencing" an external state is sufficient for "holding" it, which is a non-sequitur without a defined verification mechanism for those external states within the FP core.

3. **Causal Claim on "Progressive Disclosure":** The paper argues that "progressive disclosure" (Section 2.1) reduces token overhead and improves safety. The logical support for the safety claim is weak. The text states that revealing full schemas only after authorization "keeps the default interaction surface safer." It does not explicitly explain *why* hiding the schema prevents a safety violation (e.g., prompt injection or tool misuse) if the agent eventually receives the full schema to execute the task. The causal link between "hiding metadata initially" and "increased safety" is asserted but not logically derived from the stated premises of the protocol's design.

These issues do not invalidate the core proposal but require clarification to ensure the conclusions strictly follow from the stated mechanisms.

- [writing] In Section 2.3, the list of primitives includes a fragment starting with 'and Economic primitives'. Remove the leading 'and' to restore logical parallelism in the enumeration.
- [science] Section 2.1 claims entities can 'hold assets' and ensure 'accountability', yet Section 2.4 states FP is 'ledger-agnostic' with no native asset tracking. The logical gap: how does referencing an external payment rail enforce asset holding or prevent double-

spending without a defined verification mechanism in the core?

- [science] Section 2.1 argues 'progressive disclosure' improves safety by hiding schemas initially. The causal link is missing: if the agent eventually receives the full schema to execute, how does initial hiding prevent prompt injection or misuse? The mechanism is asserted but not derived.

paper_reviewer_overreach

Verdict: minor_revision

The manuscript makes several claims that extend beyond the evidence provided in the text, particularly regarding performance guarantees and the efficacy of the proposed governance mechanisms.

First, in the Appendix under "Payment and settlement," the authors state that in escrow mode, "the protocol itself guarantees atomicity." This is a strong technical claim. However, the description only mentions that the arbiter "freezes funds" and "transfers the funds" upon settlement. It does not describe the underlying mechanism (e.g., smart contract logic, multi-signature escrow, or a specific consensus protocol) that prevents race conditions or ensures the transfer occurs exactly once upon the correct trigger. Without detailing the atomicity mechanism, this claim is unsupported and risks overpromising on the protocol's security properties.

Second, the paper asserts that the "progressive disclosure" design principle "reduces token usage" (Section 2.2, "Entity & Trust Plane"). While this is a plausible design goal, the paper presents it as a realized benefit without providing any comparative data, benchmarks, or theoretical analysis against the "common pattern of copying full tool descriptions." As this is a white paper describing a protocol, such performance claims should be framed as hypotheses or design intentions rather than established facts until validated by the reference implementation or external benchmarks.

Finally, the manuscript claims that FP "turns social governance into a protocol-level capability" and can address issues like "manipulation, spam, and instruction injection" (Section 3.1). While the protocol defines primitives for moderation roles and policy hooks, the paper offers no analysis, simulation, or case study demonstrating that these primitives effectively mitigate these specific threats. The claim that the protocol *solves* or *addresses* these complex social engineering problems is an overreach given the current scope, which is limited to defining the data structures and message flows. The text should clarify that these are *enablers* for governance, not guaranteed solutions to the underlying social problems.

- [science] The claim that FP 'guarantees atomicity' in escrow mode (Appendix, 'Payment and settlement') overreaches. The text admits the arbiter 'freezes funds' but does not specify the cryptographic or consensus mechanism ensuring atomic transfer upon settlement. Without this, the claim of atomicity is unsupported by the provided data.
- [science] The assertion that FP 'reduces token usage' via progressive disclosure (Section

2.2) is a performance claim lacking empirical validation. The paper offers no benchmark comparing token counts or latency against existing patterns (e.g., full tool description injection). This should be qualified as a design hypothesis rather than a demonstrated outcome.

- [science] The paper claims FP makes 'social governance into a protocol-level capability' (Section 3.1) but provides no evidence that the proposed primitives (moderation roles, policy hooks) effectively prevent manipulation or spam. This is a functional claim about system behavior that requires either a threat model analysis or empirical demonstration to be supported.

paper_reviewer_safety_ethics

Verdict: minor_revision

The manuscript proposes the Foundation Protocol (FP) as a coordination layer for an emerging "agentic society," explicitly positioning safety, accountability, and governance as first-class protocol concerns (Abstract, Sec 1). This is a positive and necessary framing for a system that enables autonomous agents to hold credentials, execute code, and transact value. The paper correctly identifies that as agents move from tools to actors, the protocol layer itself must become a safety boundary (Sec 1).

However, the review identifies a significant gap in the explicit treatment of dual-use risks and adversarial scenarios. While the paper mentions "safety reports" and "dispute escalation" (Sec 4.4) and cites external security research (Sec 1), it lacks a dedicated section or detailed analysis of how FP specifically mitigates high-stakes risks inherent to its design. For instance, the ability for agents to form "AI organizations" that "hire external services" and "compete for resources" (Sec 1) creates a vector for autonomous coordination of malicious activities (e.g., distributed denial-of-service attacks, coordinated disinformation campaigns, or automated financial manipulation). The manuscript does not currently articulate specific failure modes where the protocol's "progressive disclosure" or "trust signals" could be exploited to bypass human oversight or facilitate such harms.

Furthermore, the reference implementation's reliance on "friend-list-based access control" and "social-graph gating" (Appendix, Sec F.1) raises concerns about the potential for creating isolated, unmoderated agent networks. The paper should explicitly address how the protocol prevents the formation of "echo chambers" of autonomous agents that reinforce harmful behaviors or evade external audit, particularly given the "tree topology" described in the implementation (Appendix, Sec F.3).

Finally, while the "Regulation & Oversight Plane" is described as a place for "policy evaluation" (Sec 4.4), the paper does not provide concrete examples of how this plane would handle specific adversarial inputs, such as "instruction injection" or "prompt hacking," which are known vulnerabilities in agentic systems (Sec 4.1). The current description is too abstract to assure readers that the protocol can effectively enforce safety constraints against determined adversaries.

To address these concerns, the authors should add a dedicated "Safety and Ethics" section that: 1. Explicitly lists potential dual-use risks and adversarial scenarios specific to FP's capabilities. 2. Details how the "Regulation & Oversight Plane" and "checkpoint pipeline" are designed to mitigate these specific risks. 3. Discusses the limitations of "friend-list" based access control and proposes additional safeguards against the formation of unmoderated agent networks. 4. Provides concrete examples of policy enforcement against known agent attack vectors.

Without these additions, the paper's claims regarding the safety and governability of the proposed system remain insufficiently supported.

- [writing] The manuscript describes a protocol enabling autonomous agents to execute code, manage funds, and form organizations (Sec 1, Sec 4). It lacks a dedicated 'Safety and Ethics' section explicitly addressing dual-use risks, such as agents coordinating malicious campaigns, bypassing human oversight, or facilitating financial fraud. A discussion of these risks and proposed mitigations is required.
- [writing] The reference implementation (Appendix) mentions 'friend-list-based access control' and 'social-graph gating' (App. F.1). The paper must clarify how this mechanism prevents the formation of closed, unmoderated agent networks that could amplify disinformation or coordinate harmful activities without human intervention.
- [writing] While the paper cites Microsoft's security research on agent risks (Sec 1), it does not detail how the 'Regulation & Oversight Plane' specifically handles adversarial inputs or 'instruction injection' (Sec 4.1) beyond generic policy hooks. Specific examples of policy enforcement against known agent attack vectors are needed.

paper_reviewer_scientific_evidence

Verdict: minor_revision

The manuscript proposes a novel coordination protocol (FP) for agentic societies but currently functions as a design specification and white paper rather than an empirical scientific study. From a scientific evidence perspective, the central claims regarding the protocol's efficacy, scalability, and economic advantages are unsupported by quantitative data.

The primary gap is the absence of experimental validation. Section 1.3 claims FP reduces token overhead via "progressive disclosure," yet no measurements are provided to quantify this reduction compared to standard prompt-stuffing methods. Similarly, the architecture's ability to handle "large networks" (Section 2.2) is asserted without stress-testing results (e.g., latency under load, throughput limits, or failure rates in the tree topology described in Appendix A.3). Without benchmarks, the claim that FP is a superior "foundation layer" remains theoretical.

Furthermore, the "AI company" scenario in Section 3.2 serves as a narrative proof-of-concept but lacks the rigor of a controlled experiment. It does not demonstrate how the protocol handles edge cases, such as malicious actors, network partitions, or economic

disputes, which are critical for validating the "Regulation & Oversight" plane. The reference implementation (Appendix A) is described architecturally but not evaluated; there is no data on its performance, security robustness, or adoption feasibility.

Finally, the reputation system described in Appendix A.4 introduces a five-dimensional scoring model. However, the paper does not provide the mathematical formulation for the "confidence" and "recency" weighting factors, nor does it offer evidence that this specific aggregation method prevents gaming or accurately reflects vendor quality. To move from a proposal to a scientifically grounded contribution, the authors must supplement the architectural description with empirical results from simulations or pilot deployments that validate the protocol's performance and safety claims.

- [science] The paper presents a protocol architecture and a conceptual scenario but lacks empirical validation. To support claims of 'reduced overhead' (Section 1.3) and 'scalability,' include quantitative benchmarks comparing FP against existing protocols (e.g., MCP, A2A) on metrics like token usage, latency, and message complexity.
- [science] The 'AI company' scenario (Section 3.2) is illustrative but not evidence. Provide a pilot study or simulation results demonstrating the protocol's ability to handle multi-agent coordination, dispute resolution, or economic settlement in a realistic setting with defined failure modes.
- [science] The reputation system (Appendix A.4) relies on 'confidence' and 'recency' factors without defining the underlying statistical model or aggregation function. Specify the mathematical formulation and provide evidence (e.g., sensitivity analysis) that the scoring mechanism is robust to manipulation or sparse data.

paper_reviewer_statistical_analysis

Verdict: minor_revision

The manuscript "Foundation Protocol: A Coordination Layer for Agentic Society" is primarily a systems architecture and protocol design paper. Consequently, it does not contain traditional statistical analyses, hypothesis testing, or empirical datasets requiring standard statistical review (e.g., t-tests, ANOVA, regression models). The "science" presented is architectural and conceptual rather than empirical.

However, the Appendix (Section A.2.4, "Reputation") introduces a quantitative reputation scoring system derived from contract data. This section currently lacks the necessary statistical rigor for reproducibility. The authors describe a five-dimensional profile (Quality, Reliability, Collaboration, Efficiency, Integrity) weighted by "confidence" and "recency" factors. Without specifying the underlying distribution of the data, the sample size (N) of contracts used to calibrate these weights, or the method for calculating the "confidence factor" (e.g., Wilson score interval, Bayesian priors), the metric remains a black box. To meet the standards of a scientific protocol specification, the authors should provide the mathematical formulation for the aggregation function and, if possible, a sensitivity analysis or confidence intervals for the resulting scores.

Additionally, the historical claims in Section 1.1 regarding "intelligence density" (Table 1) are qualitative. If the authors intend to imply a quantitative trend or rely on specific economic data to support the "step change" assertions, they should cite the specific datasets and statistical summaries used. As it stands, the lack of statistical detail in the reputation mechanism is the only area where the analysis falls short of reproducibility standards.

- [science] The manuscript describes a 'five-dimensional' reputation profile (Quality, Reliability, Collaboration, Efficiency, Integrity) in Appendix A.2.4. However, it lacks statistical validation of this metric. Please report the sample size (N) of contracts used to derive these scores, the specific aggregation method (e.g., weighted mean, Bayesian smoothing), and confidence intervals or error margins for the reported dimensions to ensure reproducibility.
- [writing] Section 1.1 and Table 1 present a historical narrative of 'intelligence density' across industrial revolutions. While conceptual, if any quantitative data (e.g., productivity metrics, R&D spend) was used to construct these claims or the table, the statistical sources, normalization methods, and uncertainty estimates must be explicitly cited and described.

paper_reviewer_writing_quality

Verdict: minor_revision

The manuscript presents a sophisticated and ambitious vision for a coordination layer in agentic societies. The writing is generally high-quality, characterized by a clear, authoritative tone and well-structured arguments. The authors successfully navigate complex conceptual ground, moving from historical analogies to specific protocol design without losing the reader. The prose is dense but readable, and the logical flow between sections is strong.

However, there are a few specific instances of mechanical errors that disrupt the reading experience and suggest a need for a final proofread. Most notably, in the Introduction (Section 1), the text lists existing protocols and includes a sentence fragment: "...among agents in open networks [anp_paper]. and UCP targets commerce..." The period before "and" incorrectly breaks the sentence structure. A similar error appears in Section 3.3, where a sentence begins with a lowercase "and" following a period: "...as first-class objects. and Economic primitives standardize..." These are likely copy-paste or editing artifacts that should be corrected to maintain the professional polish of the rest of the document.

Additionally, there is a minor spacing issue in Section 1.1 where a citation bracket is attached directly to the preceding word ("further[ref]"). While this is a minor typographical detail, correcting it improves the overall visual cleanliness of the text.

Overall, the writing is compelling and effectively communicates the necessity and design of the Foundation Protocol. Addressing these few grammatical and punctuation slips will ensure the manuscript is as polished as its technical content.

- [writing] In Section 1 (Introduction), the sentence 'ANP emphasizes discovery and ne-

gotiation among agents in open networks [anp_paper]. and UCP targets commerce...' contains a grammatical error where a period precedes the conjunction 'and'. This creates a sentence fragment. Please merge this into the preceding sentence or remove the period.

- [writing] In Section 3.3 (Interaction @ Organization Plane), the sentence 'Sessions and organizations capture groups, roles, membership, and delegation as first-class objects. and Economic primitives standardize...' repeats the error of starting a sentence with a lowercase 'and' following a period. Please correct the punctuation and capitalization to ensure flow.
- [writing] In Section 1.1, the phrase 'Compatibility and network effects then accelerated this process further[ref]' lacks a space between the word 'further' and the citation bracket. Please insert a space for proper typesetting and readability.